MCO-1D PRE-FLIGHT CHECKLIST

Run before every flight — do not skip

1. PHYSICAL INSPECTION

- ■ **All 8 wings at 60° sweep, no binding**
- ■ All 16 servo linkages seated and locked
- ■ **All 8 ropes in groove — no fraying visible**
- ■ 8 encoder magnets secure, no wobble
- ■ CG balance: aircraft hangs level ±5°
- ■ No loose screws, bolts, or connectors
- ■ **Flight battery secured, connector locked**
- ■ **Servo LiPo: charged, above 7.2V**

2. ELECTRONICS

- ■ 8×ODrive S1 power lights: solid green
- ■ **Pixhawk GPS: solid blue, ≥6 satellites**
- ■ Pitot tube: forward-facing, unobstructed
- ■ RC receiver: binding confirmed
- ■ **RC range check: all sticks respond in GS**
- ■ **Lever: forward = more thrust confirmed**

3. SOFTWARE

- ■ **Pi booted: mco_main.py running**
- ■ GS laptop connected: port 5757
- ■ **persist_load_all: learning restored**
- ■ soft_stop_status: all 8 wings PASS
- ■ cg_status: stability margin positive
- ■ **failure_briefing: 0 Catastrophic**
- ■ **fault_summary: no active faults**
- ■ gps_ok=True, pitot valid in telemetry
- ■ inbox_check: no pending imports

4. BEFORE ARM

- ■ **Area clear: 10m radius, observers back**
- ■ Wind < 8 m/s, no precipitation
- ■ **Home position set at ARM — do not move**
- ■ **Emergency procedures briefed to all**
- ■ AI GS: ws://localhost:5758 connected

ARM: type `arm` in GS or say "arm the aircraft" to AI assistant

MCO-1D IN-FLIGHT EMERGENCY CARD

Keep visible during all flights

WING FAILURE — PROGRESSIVE

- 6–7 wings: degraded warning, continues
- 5 wings: advisory to land, may continue
- **≤4 wings: LAND NOW auto-engages**
- Lever: descent rate (50%=0.20m/s, 100%=0.08m/s)
- **Pilot RETAINS full roll/yaw/pitch authority**
- **Steer to clear area — do not fight descent**

RC LINK LOSS

- Aircraft climbs 30m, orbits, waits 10s
- **Link not restored: RTH to ARM position**
- **RTH descends 0.3m/s — clear landing zone**
- Do not move from ARM position during flight

STALL WARNING

- stall_detector_status > 0.45 alert
- **Increase lever — do NOT reduce**
- Reduce bank, allow nose to drop

IMMEDIATE ACTIONS — ANY EMERGENCY

- **Call CLEAR — warn all observers**
- **Maintain visual contact at all times**
- Lever to 80%+ unless instructed otherwise

EMERGENCY STOP — LAST RESORT ONLY

- **GS: emergency_stop OR panel: E-STOP**
- **All motors stop immediately — aircraft drops**
- **ONLY if crash certain AND people at risk**
- **Never use over people or near crew**

LAND_NOW FROM THE GROUND

- Wing frequency audibly reduces near ground
- Descent: 0.08–0.50 m/s depending on lever
- May drift with wind — steer gently
- **Full pilot authority throughout**
- Touchdown < 0.08m/s — expect soft landing
- **After: lever 0%, then GS: disarm**

KEY EMERGENCY COMMANDS

- fault_summary — what is wrong
- controlled_landing_status — land_now state
- fault_wing_restore W1 — after repair

MCO-1D POST-FLIGHT LEARNING CARD

Run after every flight — do not skip

STEP 1 — SAFE FIRST

- ■ Lever to 0%
- ■ **GS: disarm**
- ■ **Confirm all wings stopped**
- ■ Disconnect flight battery
- ■ Fold wings for storage

STEP 2 — SAVE LEARNING (BEFORE POWER OFF)

- ■ **GS or AI: persist_checkpoint**
- ■ Wait: "Learning persisted — N components"
- ■ **GS: knowledge_export_now**
- ■ Files in /home/pi/knowledge/outbox/
- ■ **Copy outbox to USB or laptop**

STEP 3 — RECORD OBSERVATIONS

- ■ GS: gs_inject_note "observation"
- ■ Record anything unusual or interesting
- ■ "Left drift in hover — wind from NE"
- ■ "Dart sluggish below 8m/s"
- ■ "H-PH-2 needs calm-day retest"

STEP 4 — REVIEW LEARNING

- ■ **GS: learning_history**
- ■ What improved and by how much this flight
- ■ GS: mechanism_weight_report
- ■ Which mechanisms rank highest now
- ■ GS: hypothesis_results
- ■ Which research hypotheses were promoted

STEP 5 — FLEET SHARING

- ■ Each drone: knowledge_export_now
- ■ Collect _fleet.json files to one laptop
- ■ **merge_fleet_knowledge d1.json d2.json**
- ■ **knowledge_import merged.json on each drone**
- ■ **All drones start from combined learning**

STEP 6 — INJECT IDEAS

- ■ **gs_inject_hypothesis name="idea" param=val**
- ■ Queues for next discovery session
- ■ Or tell AI assistant in plain English

SHUTDOWN: power off Pi ONLY AFTER persist_checkpoint

MCO-1D ASSEMBLY SEQUENCE CARD

First build — follow in order, verify each step

PHASE A — SERVOS (DO FIRST)

- ■ **Buy: 18×AGFRC IB53BHU + AGF-SPV3 programmer**
- ■ **Programme ALL 16 servos to 170° max travel**
- ■ ADC saturates at 180° — 170° gives 10° margin
- ■ Label: W1p,W1s,W2p…W4s
- ■ Install servos — do not connect ropes yet

PHASE B — DRIVES AND ENCODERS

- ■ **Mount 8×ODrive S1 (node_id 0–7)**
- ■ CAN-FD bus: daisy-chain all 8, 120Ω terminators
- ■ **Install encoder magnet on each shaft**
- ■ **Verify: position reads full arc smoothly**
- ■ **Run mco_odrive_setup.py — sector arc limits**
- ■ GS: soft_stop_status — all 8 PASS

PHASE C — ROPES

- ■ Thread rope through each sector wheel
- ■ **FATIGUE TEST: drive to failure, record I²×cycles**
- ■ **Update ROPE_LIFE_I2_CYCLES in code**
- ■ Connect ropes, check tension
- ■ **No fraying at reversals under power**

PHASE D — MEASUREMENTS

- ■ **Run python3 mco_build_measurements.py**
- ■ **A1: actual MTOW — weigh complete aircraft**
- ■ C1: gear ratio — count teeth
- ■ C2/C3: sector arc angles — protractor
- ■ **K_PITCH_MM_PER_DEG: tilt 5°, measure CG shift**

PHASE E — ELECTRONICS AND RC

- ■ Pixhawk: GPS clear sky view
- ■ **Pitot: forward-facing, no bends**
- ■ Wire MCP3208 ADC for servo feedback
- ■ **Run mco_rc_verify.py — confirm channels**
- ■ **Lever: forward = more power (FW convention)**
- ■ Rudder: right pedal = right yaw

PHASE F — SOFTWARE COMMISSIONING

- ■ **Boot Pi: mco_main.py, no errors**
- ■ Laptop: python3 mco_ws_bridge.py [pi_ip]
- ■ Open AI ground station in browser
- ■ **TC-01 bench: tc01_start**
- ■ **TC-02 captive hover: tc02_start**
- ■ **TC-03 truck FW: tc03_start**
- ■ **TC-04 first free flight: tc04_start**

MCO-1D SAR OPERATIONS CARD

Search and Rescue mission procedures

TRANSIT TO SCENE

- Climb 30m+, lever to 70%
- Airspeed_hold engages after 1.5s
- Monitor: V_ms, h_agl, fault_summary

APPROACH AND SITE HOLD

- Reduce lever: blend zone 3–5 m/s
- **Below 3 m/s: hover auto-engages**
- **Crew hover ON: stick = velocity**
- Stick fwd = commanded forward speed
- **Release stick: decelerates, stops**
- **GS or AI: position_hold_engage**
- **Aircraft holds GPS — hands off sticks**

CREW OPERATIONS

- Pilot monitors altitude and drift
- **Reposition: apply stick, aircraft moves**
- Release: re-locks at new position
- Altitude: lever above/below 50%

LOITER / ENDURANCE

- **Extended on-station: GS h36_engage**
- H-36: minimum power draw
- **RTH with min 20% battery reserve**
- h36_loiter_report: time remaining

CREW HOVER — HOW IT WORKS

- **Default ON — no switch needed**
- **Stick centred = 0 m/s = holds position**
- Small stick = slow creep to target
- Large stick: 5m/s fwd, 3m/s lateral max
- Classic feel: crew_hover_disable

RETURN TO HOME

- **AI GS: "return to home"**
- Climbs, tracks to ARM position
- **Descends 0.3m/s — clear landing zone**
- **Manual override: take stick any time**
- After: disarm, post-flight card

DART — RAPID HEADING CHANGE

- **Lateral joystick + SAME-SIDE rudder**
- Slow = coordinated turn; fast = dart
- Release either = smooth exit
- Pitch axis adds vertical component